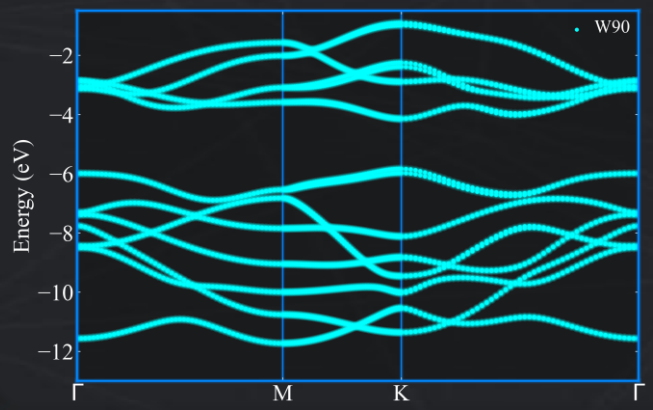
WannierBSE

Zero-Effort Setup: Direct Integration with Wannier90

Connect your Wannier90 results, tune your simulation parameters, and power your discovery with *WBSE.m*

1 Bring your Wannier90 files

wannier90.win
wannier90_*.xsf
wannier90_hr.dat



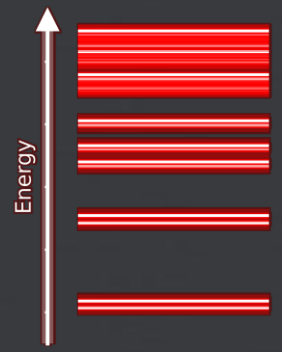
2 Set Simulation Parameters

- control.txt
- exchange_control.txt
- kmesh_control.txt
- WTB_control.txt
- WF_centers.txt
- dielectric_control.txt

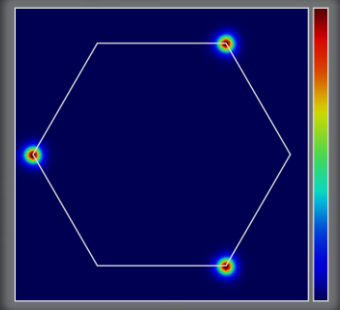
Note: Requires MATLAB Parallel Computing Toolbox

3 Run *WBSE.m*

Exciton Energy Spectrum



Exciton Wave Functions

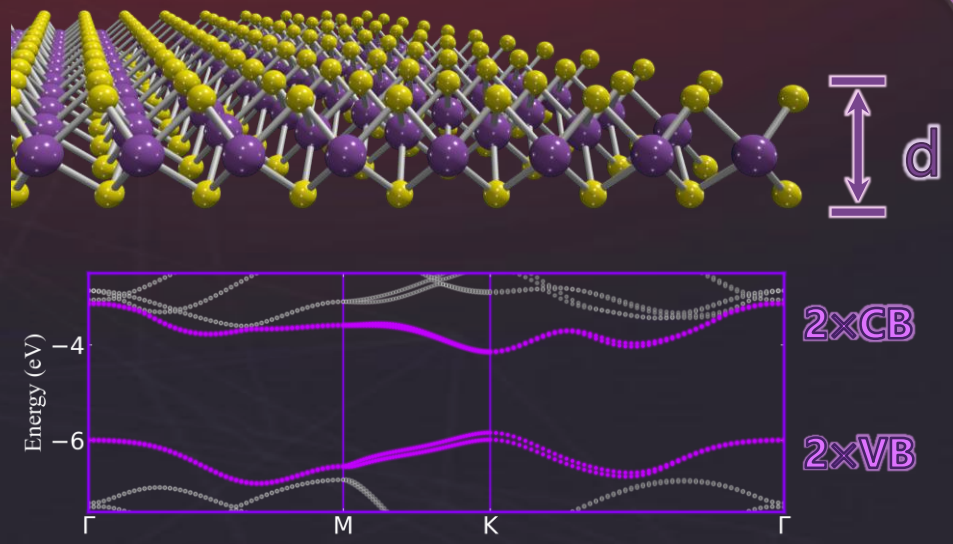


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WannierBSE

Calculation Parameters & Parallel Scaling

Define the computational settings for the BSE calculation via control.txt



Requires MATLAB Parallel Computing Toolbox

control.txt

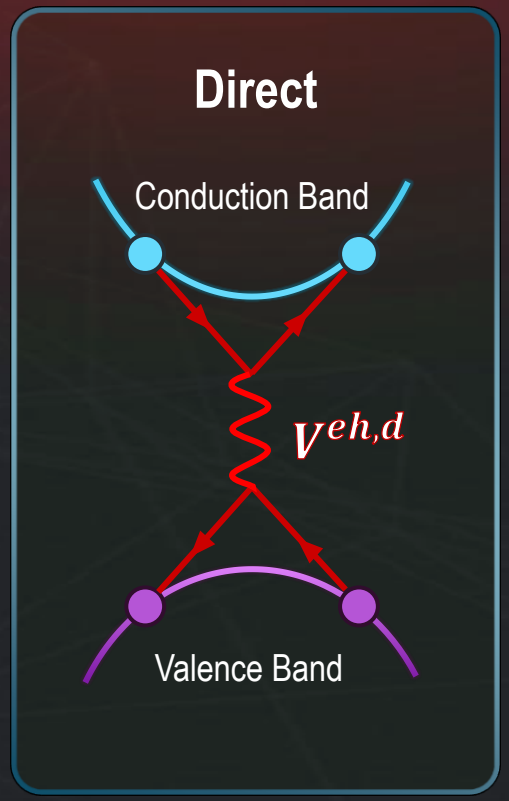
```
d = 6.26          % 2D material thickness in angstroms
G_dir_max = 1     % Direct-interaction G cutoff
%--- BSE calculation options ----
NVAL = 100        % # of exciton eigenvalues to find
Nc = 2            % Conduction bands in BSE
Nv = 2            % Valence bands in BSE
%--- Parallel calculation parameter ---
Npar_TB   = 12    % # of cores for TB band generation
Npar_Xi   = 12    % # of cores for SR Xi integrals
Npar_HBSE = 12    % # of cores for BSE construction
```

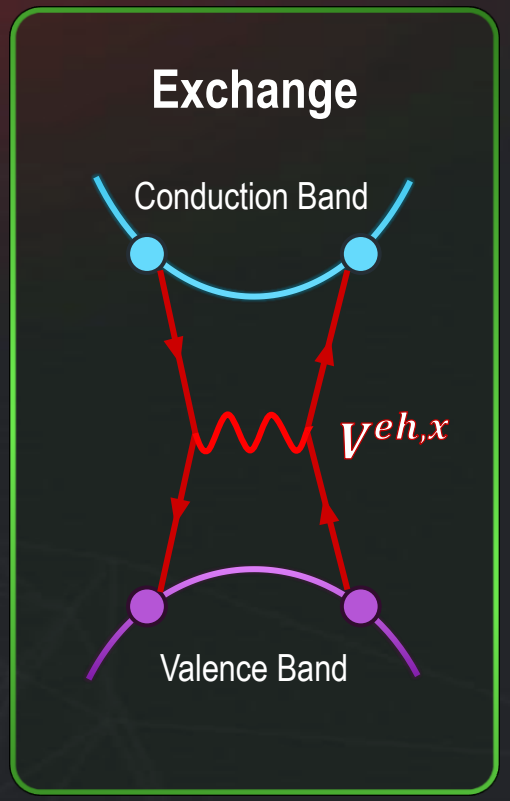
Input monolayer thickness (d), select your active band window (Nc, Nv), and set your target eigenvalues. Accelerate each calculation stage through parallel processing

WannierBSE

Configure the Short-Range Exchange Interaction

Enable short-range exchange through exchange_control.txt





exchange_control.txt

```
Exchange_Interaction    = true   % Enable SR exchange (G~=0)
Use_Precomputed_Chi  = true  % Load precomputed chi
cleanup_xsf                 = false  % Keep raw XSF files

% --- Exchange Parameters ---
M_max                 = 1      % Maximum neighbor shell
G_max                 = 4.2    % Exchange G-vector cutoff
tol_probability       = 0.98   % Wannier-box probability cutoff
M_max_tolerance  = 0.01 % Neighbor-shell distance tolerance
nn_distance_tolerance = 0.15  % NN classification tolerance
```

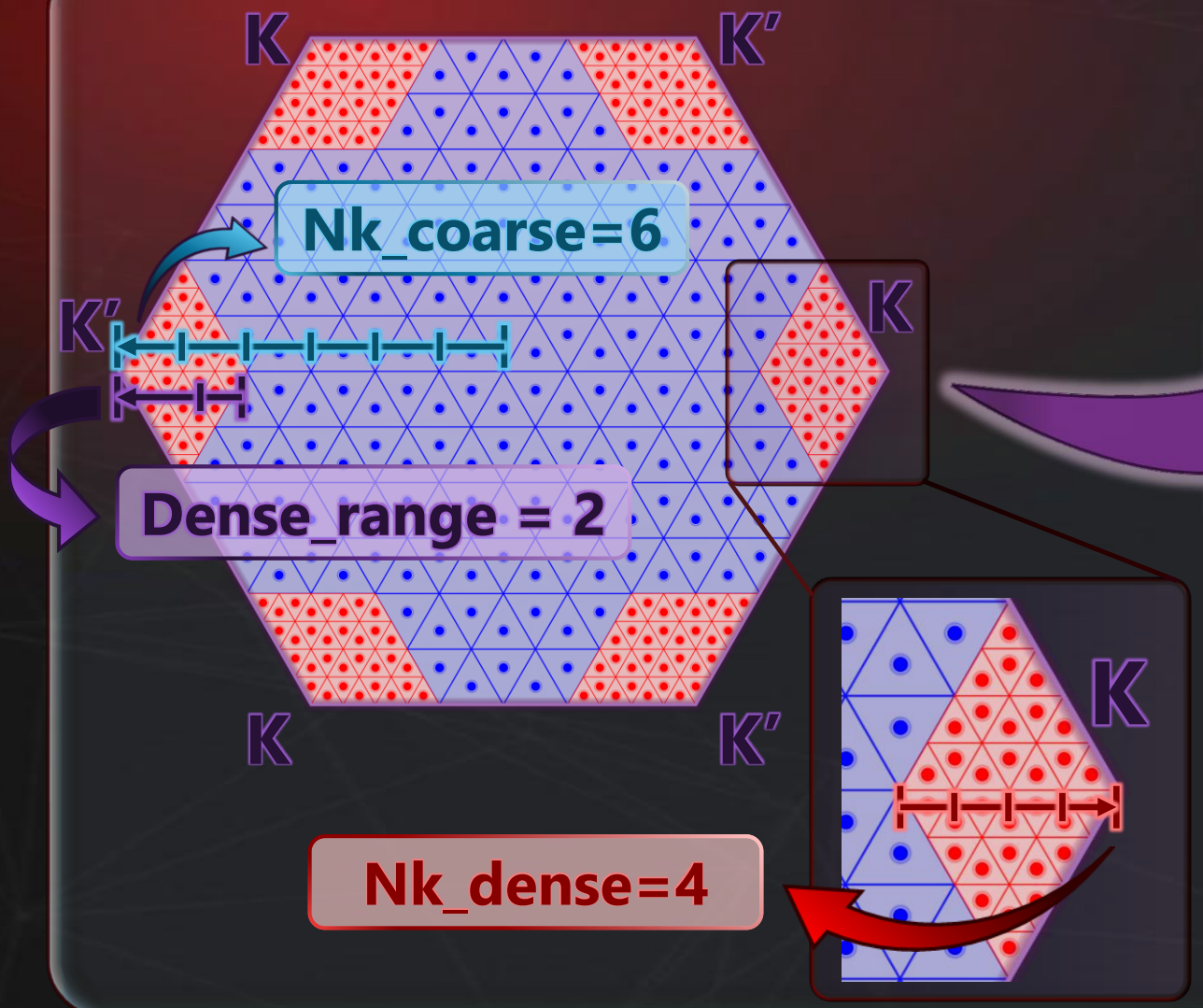
Enable exchange interaction
Set Exchange_Interaction = true

Optionally reuse precomputed chi
Set Use_Precomputed_Chi = true

Keep XSF files
Set cleanup_xsf = false

WannierBSE

Set your non-uniform k-mesh



kmesh_control.txt

```
% ---  Control parameters for k-mesh generation  ---

Nk_coarse = 6          % K-points per direction in coarse region

Nk_dense = 4           % K-points per direction in dense region

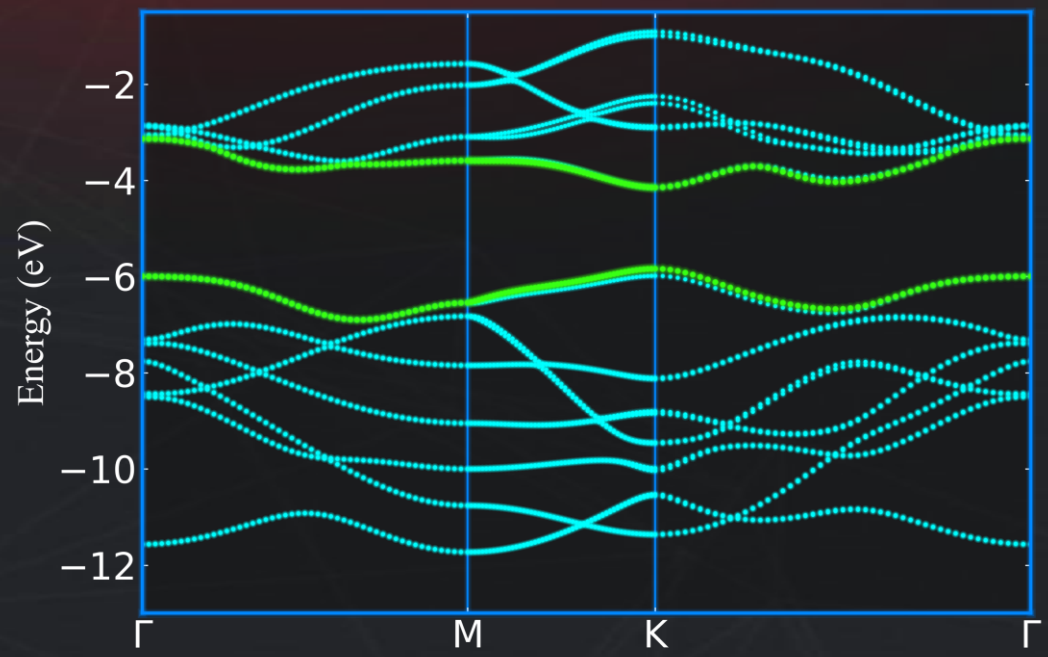
Dense_range = 2 % Dense-region size in coarse-grid units
```

WannierBSE

Zero-Effort Setup: Direct Integration with Wannier90

Configure your WTB_control.txt file directly from your Wannier90 results

wannier90_hr.dat



WTB_control.txt

```
%--- Control parameters for Wannier Tight-Binding

VBM = 14        % Top valence band index
CBM = 15        % Lowest conduction band index
```

Select the top valence
and lowest conduction bands
from your wannier90_hr.dat output.

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WannierBSE

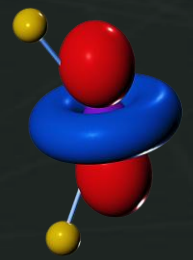
Map Wannier Functions to Atomic Centers

Assign each Wannier function to its atomic center through WF_centers.txt

wannier90.wout

Wannier centers and nearest atomic sites

Plot Wannier Functions



Visual localization verification

WF_centers.txt

% i-th Wannier Function center at τ_I atom

i	I	Spinor
1	2	up
2	2	down
3	2	up
4	2	down
5	2	up
...		

1. WF index
2. Associated atomic index
3. Spinor Label: up or down

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WannierBSE

Dielectric Screening Function

dielectric_control.txt

```
% --- Material Parameters for the TMD-ML
alpha = 1.55      % Fitting parameter
Epl   = 22.5      % Plasmon peak energy (eV)
ep_p  = 13.24     % In-plane dielectric constant

% --- Number of Layers
t_layers = 2      %  layers ABOVE the TMD-ML
s_layers = 1      %  layers BELOW the TMD-ML

% --- Dielectric Constants
t_dielectric = 1, 5.89
s_dielectric = 3.9

% --- Boundaries (Å)
layer_boundaries =103.13, 3.13, -3.13
```

